

Long Atelier Shell Soak Article

Atelier Test Suite

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Abstract

This deliberately long document exercises scrolling, wrapping, compilation, outline navigation, comments, error markers, and synchronization between the source editor and the generated PDF. The prose is repetitive by design so that the document remains safe to edit during an extended user soak.

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1 Introduction

Long-form editing behaves differently from a tiny fixture. Cursor movement, viewport restoration, line wrapping, and active-section tracking must remain stable after hundreds of lines have been visited. This article therefore contains many sections with paragraphs long enough to wrap several times.

The experimental shell should preserve the editor width while the PDF preview is resized. Closing and reopening the document should restore preferences without coupling volatile state to another open tab.

1.1 Motivation

A realistic soak needs enough material to reveal delayed layout defects. Short examples rarely expose stale measurements, misplaced gutters, or synchronization errors that only appear after scrolling far from the beginning of a document.

1.2 Objectives

The primary objective is to test ordinary work: writing text, saving changes, compiling, reading diagnostics, and moving between source and PDF. A secondary objective is to confirm that comments remain attached after paragraph reflow.

2 Background

Document editors combine several independent systems. The source surface owns text and selections, the compiler reads files from disk, the PDF surface renders pages, and SyncTeX maps positions between those representations.

Each system may finish asynchronously. A robust interface must prevent a late response from repainting a destroyed tab or replacing a more important status message with unrelated background information.

2.1 Source model

The source model stores the current text, cursor, selections, and undo history. Edits should mark the document as modified immediately. Compilation should save the current buffer before asking the backend to read the file from disk.

2.2 Preview model

The preview model may be refreshed after every successful compilation. Its width is independent from the source model and is controlled by the draggable separator in the unified shell.

3 Methods

The soak procedure begins by opening this file from the gallery with the experimental LaTeX route enabled. The user then resizes the split, edits several paragraphs, saves, compiles, and navigates through the outline.

3.1 Editing protocol

First, place the cursor in this paragraph and insert a short sentence. Save the document with the keyboard shortcut and confirm that the modified indicator returns to the saved state. Undo and redo the edit to exercise history.

Second, select a phrase and add an anchored comment. Rewrap the paragraph at a different column width, reload the tab, and verify that the comment remains on the intended phrase.

3.2 Compilation protocol

Compile while the buffer contains an unsaved edit. The shell must save before compilation, and the PDF should contain the new sentence. Repeated clicks on the compile control must start only one save-and-build chain.

3.3 Synchronization protocol

Use forward synchronization from a line near the middle of the document. Then use reverse synchronization from the PDF and confirm that the editor returns to the corresponding source line without losing the current split width.

4 Results

The following subsections provide enough vertical distance to test navigation and scrolling. They contain stable labels that can also be used for comments.

4.1 Layout stability

The unique phrase LAYOUT-STABILITY-ANCHOR belongs to this subsection. Resize the separator repeatedly and verify that neither pane collapses below a usable width. The permitted range should keep both panes visible.

4.2 Status stability

The unique phrase STATUS-STABILITY-ANCHOR belongs here. A successful compile status should remain primary even if background version persistence reports a secondary warning.

4.3 Comment stability

The unique phrase COMMENT-STABILITY-ANCHOR is intended for an anchored comment. After rewrapping this paragraph, the annotation should still identify the same text rather than an obsolete line and column pair.

4.4 Multiple tabs

Open this article beside the long report fixture. Change the wrap preference in one tab and move the cursor in the other. Volatile editor state must remain independent while intentionally shared preferences remain synchronized.

5 Extended Discussion

This section adds realistic length. An editor used for manuscripts must remain responsive after many source lines, repeated compiles, and several preview refreshes. The test is not a benchmark, but obvious pauses or layout jumps are signals worth recording during the soak.

Paragraph one discusses navigation. Moving through the outline should place the cursor at the selected heading and focus the source editor. The outline should then close so it does not obscure subsequent edits.

Paragraph two discusses diagnostics. A broken command introduced temporarily below should create a visible error marker and a clickable log entry. Remove the command afterward and compile again to ensure old markers are cleared.

Paragraph three discusses teardown. Start a compilation and immediately close the tab. Re-opening the file should not reveal stale logs, duplicate listeners, or controls that react more than once to a single click.

Paragraph four discusses persistence. The split percentage is a user preference and should survive a reload. Hiding the preview should expand the editor, while showing it again should restore the last useful percentage.

Paragraph five discusses accessibility. The separator and controls need stable labels so that keyboard and assistive-technology users can understand their purpose even when icons are used in the visible interface.

6 A Small Quantitative Example

The following equation gives the editor several mathematical lines to render:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2. \quad (1)$$

For a simple linear model,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2). \quad (2)$$

Scenario	Source lines	Expected pages
Short fixture	20	1
Long article	150+	4+
Long report	180+	5+

Table 1: Illustrative soak documents.

7 Conclusion

This long article is successful when ordinary editing remains predictable over an extended session. Record any incorrect status, misplaced annotation, broken split, stale PDF, or navigation mismatch before enabling the shell by default.

A Checklist

1. Resize the editor and PDF panes.
2. Hide and restore the split preview.
3. Edit, save, undo, and redo.
4. Compile a dirty buffer.
5. Navigate through the outline.
6. Add a comment and rewrap its paragraph.
7. Exercise forward and reverse SyncTeX.
8. Open a second LaTeX tab and compare behavior.
9. Close a tab during compilation.
10. Reload and verify persisted preferences.